

Interactive Shiny App Project

MAS 627-57 Programming for Data Analytics

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Interactive Shiny App - AACR Project GENIE

Describe the app you built

We built a Dashboard in R Shiny using the AACR Project GENIE clinical dataset, which includes over 200,000 tumor samples from cancer centers worldwide. The goal was to create a tool that goes beyond raw numbers to tell a story about the demographic background of patients, what cancers they face, and the survival outcomes after DNA sequencing.

What does it do

The dashboard makes complex patterns in the GENIE data easy to interpret. It shows which cancers are most common, how demographics like age, sex, and race vary, and survival outcomes after sequencing. Instead of overwhelming users, it highlights key trends across thousands of cases to give a clearer picture of the patient population.

Why did you choose this & why is this interesting

Cancer touches nearly everyone, making the insights broadly relatable, and for our team the choice was personal too, shaped by Oliver's work at the University of Miami Health System, which underscored the value of data-driven medicine. Project GENIE is also used in precision oncology to study tumor genetics and guide treatment, giving us hands-on experience applying data to real medical challenges.

What features does it have

- **Multiple tabs:** Dashboard, data exploration and about section.
- **KPIs:** Important and quick numbers to highlight the most relevant information.
- **Overview tab:** Patient counts, vital status breakdown, and top cancer types.
- **Survival tab:** A grouped bar plot showing years from sequencing to death (0, 1, 2, 3, 4, 5, 6+).
- **Demographics tab:** Age distributions, sex ratios, and race/ethnicity counts.
- **Filter sidebar:** Options to subset the dataset by sex, race, cancer type, and sequencing center, with all plots updating automatically.
- **ShinyApps deployment:** <https://oliversinn.shinyapps.io/genie-dashboard/>
- **GitHub with automatic deployments:** <https://github.com/Oliversinn/genie-dashboard>